

AI / Data Science

Task Report — Day 02

Diabetes Dataset · EDA · 5 Visualizations · Logistic Regression

DATE

13 June 2026

DOMAIN

Artificial Intelligence / Data Science

DATASET

Pima Indians Diabetes Dataset (768 rows x 9 cols)

1 Task Overview

#	Objective	Status
1	Clean and preprocess a real-world dataset	✓ Done
2	Perform detailed Exploratory Data Analysis	✓ Done
3	Create 5 visualizations with insights	✓ Done
4	Build a simple Machine Learning model	✓ Done
5	Document findings and model performance	✓ Done

2 Dataset — Pima Indians Diabetes

Source: plotly/datasets (GitHub) **Rows:** 768 **Columns:** 9 **Task:** Binary Classification (Diabetic vs Non-Diabetic)

Column	Description	Issue
Pregnancies	Number of pregnancies	None
Glucose	Blood sugar level	Had 0s
BloodPressure	Blood pressure reading	Had 0s
SkinThickness	Skin fold thickness	Had 0s
Insulin	Insulin level	Had 0s
BMI	Body mass index	Had 0s
DiabetesPedigreeFunction	Genetic diabetes risk score	None
Age	Patient age in years	None
Outcome	0 = No Diabetes, 1 = Diabetes (Target)	—

3 Data Cleaning & Preprocessing

The dataset showed zero missing values initially — but five columns contained biologically impossible 0 values (a person cannot have 0 glucose or 0 BMI). These were treated as hidden missing values and replaced with each column's median.

Column	Zero Values Found	Action
Glucose	5	Replaced with median
BloodPressure	35	Replaced with median
SkinThickness	227	Replaced with median
Insulin	374	Replaced with median
BMI	11	Replaced with median

```
zero_cols = ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']
df[zero_cols] = df[zero_cols].replace(0, np.nan)
df[zero_cols] = df[zero_cols].fillna(df[zero_cols].median())
```

4 Feature Engineering

Two new columns were created to enable more meaningful analysis:

New Column	Values	Purpose
AgeGroup	Under 30 / 30-45 / 45-60 / Over 60	Group patients by life stage
BMICategory	Underweight / Normal / Overweight / Obese	Standard medical BMI bands

```
df['AgeGroup'] = pd.cut(df['Age'], bins=[0,30,45,60,100],
                        labels=['Under 30', '30-45', '45-60', 'Over 60'])
df['BMICategory'] = pd.cut(df['BMI'], bins=[0,18.5,24.9,29.9,100],
                           labels=['Underweight', 'Normal', 'Overweight', 'Obese'])
```

5 Visualizations

Chart 1

Outcome Distribution (Bar Chart) → chart1_outcome_distribution.png

Insight

500 patients (65.1%) have no diabetes vs 268 (34.9%) who do. The dataset is moderately imbalanced.

Chart 2

Glucose by Outcome (Histogram) → chart2_glucose_distribution.png

Insight

Diabetic patients show significantly higher glucose levels (mostly above 120) compared to non-diabetic patients who cluster around 90-110.

Chart 3

BMI by Outcome (Box Plot) → chart3_bmi_boxplot.png

Insight

Diabetic patients have a higher median BMI (~35) vs non-diabetic (~30), suggesting obesity is a meaningful risk factor.

Chart 4

Diabetes Rate by Age Group (Bar Chart) → chart4_diabetes_by_age.png

Insight

Diabetes rate rises from ~28% in the under-30 group to over 55% in the over-60 group — risk increases significantly with age.

Chart 5

Correlation Heatmap (Heatmap) → chart5_correlation_heatmap.png

Insight

Glucose has the highest correlation with Outcome (0.49). BMI and Age also show meaningful positive correlations.

6 Machine Learning Model

Model Configuration

Setting	Value
Algorithm	Logistic Regression
Features Used	Glucose, BMI, Age, Insulin, BloodPressure, Pregnancies
Target	Outcome (0 = No Diabetes, 1 = Diabetes)
Train Split	80% (614 samples)
Test Split	20% (154 samples)
Scaling	StandardScaler applied before training
Accuracy	~77%

Model Code

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42)
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print('Accuracy:', round(accuracy_score(y_test, y_pred)*100, 2), '%')
```

Model Performance

Class	Precision	Recall	F1-Score
No Diabetes	0.81	0.86	0.83
Diabetes	0.69	0.62	0.65
Overall Accuracy	—	—	~77%

7 Key Findings & Conclusion

- Glucose is the single strongest predictor of diabetes (correlation: 0.49)
- 374 out of 768 insulin values were hidden zeros — careful cleaning was essential
- Diabetic patients had notably higher BMI (~35) vs non-diabetic (~30)
- Diabetes risk grows sharply with age — over 55% rate in the 60+ group
- Logistic Regression achieved ~77% accuracy on unseen test data
- The model predicts non-diabetic cases more reliably than diabetic ones

Task Status: Complete — All 5 objectives met. Dataset cleaned, 5 visualizations created with written insights, and a Logistic Regression classifier built and evaluated. Project is structured, documented, and pushed to GitHub.